

Contents



MARKET SURVEY

94 How To Lead The LED Lighting Boom In India

EDITOR : Ramesh Chopra

EDITORIAL : Editorial Secretary
CORRESPONDENCE : Phone: 011-26810601; E-mail: editsec@efy.in
(Technical queries: efylab@efy.in)

SUBSCRIPTIONS & : Phone: 011-26810601 or 02 or 03
MISSING ISSUES : E-mail: support@efy.in

BACK ISSUES, : Kits'n'Spares, New Delhi
BOOKS, CDs, : Phone: 011-26371661, 26371662
PCBs etc. : E-mail: info@kitsnspares.com

NEWSSTAND : Ph: 011-40596600, 8800295612
DISTRIBUTION : E-mail: efycirc@efy.in

ADVERTISEMENTS : Ph: 011-26810601 or 02 or 03
NEW DELHI : E-mail: efyenq@efy.in
(HEAD OFFICE)

MUMBAI : Ph: 022-24950047, 24928520
E-mail: efymum@efy.in

BENGALURU : Ph: 080-25260394, 25260023
E-mail: efylbr@efy.in

PUNE : Ph: 08800295610, 09870682995
E-mail: efypune@efy.in

GUJARAT : Ph: 079-61344948
E-mail: efyahd@efy.in

JAPAN : Tandem Inc., Ph: 81-3-3541-4166
E-mail: japan@efy.in

SINGAPORE : Publicitas Singapore Pte Ltd
Ph: +65-6836 2272
E-mail: singapore@efy.in

TAIWAN : J.K. Media, Ph: 886-2-87726780 ext. 10
E-mail: taiwan@efy.in

UNITED STATES : E & Tech Media
Ph: +1 860 536 6677
E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at International Print-o-Pack Ltd, C-4 to C-11, Hosiery Complex, Phase-II Extension, NOIDA-201305, Gautam Budh Nagar, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2019. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period Year	Newstand Price (₹)	You Pay (₹)	Overseas Digital	Overseas Print
Five	4800	2880	US\$ 44	—
Three	2880	2015	US\$ 31	—
One	960	770	US\$ 12	US\$ 100

Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com>

SMART WORLD

- 16** SOFTWARE: Business Intelligence Tools That Enhance Productivity
- 22** SECURITY: Karen Ravindranath, director - IoT, WebNMS: "For any business, having an internal team for the IoT security is compulsory"
- 24** GREEN TECH: Building Environment-Friendly Spaces

STARTUP ZONE

- 98** An Amalgamation of Robotics and AI
- 99** Electric Bikes Changing Perceptions
- 100** A Cellphone that Can be Worn Like a Ring
- 100** FUNDING: Somshubhro (Som) Pal Choudhury, partner, Bharat Innovation Fund: "Look for opportunities to partner rather than build the entire solution on your own"

EVERYTHING YOU WANTED TO KNOW ABOUT...

- 28** INTERNET OF THINGS: Smart Cities and the IoT: A Relationship to Build the Future
- 30** TECH FOCUS: A Guide for Building Sustainable Smart Cities
- 38** HEALTHCARE: Building the Foundation for Harmless Pest Control using Electronics
- 42** ARTIFICIAL INTELLIGENCE: AI in e-Waste Management: Dig Gold and Burn the Shit!
- 48** INTERNET: Communication and Internet Technology: Interconnecting Networks (Part 2 of 6)

DESIGN

- 55** EMBEDDED: New Chips on the Block Making SoCs Outdated
- 58** RESEARCH & DEVELOPMENT: Seeing Beyond Darkness with Smart Vision Wallet
- 60** DESIGN: Design Considerations for Autonomous Vehicles
- 64** MOBILITY: Writing the Rules for Autonomous Vehicles
- 66** TEST & MEASUREMENT: Various Tools Needed for T&M of Displays

INTERVIEW

- 69** Tanay Krishna, marketing head, Centre of Development of Telematics (C-DOT): "Standardisation, if not taken seriously, can lead to serious security threats"
- 70** Tasleem Arif, vice president and head - R&D, OPPO India: "You cannot develop, invent or create anything in isolation"
- 72** Atul Gupta, founder and chief executive officer, InSync Tech-Fin Solutions: "iPaaS connects discrete applications to work as one, thus saving effort and directly impacting the operational cost"

SOFTWARE

- 105** Software for a Smart Home
- 106** Mobile Apps for Smart Living

eSTYLE

- 107** FIRST LOOK: Some New Gadgets
- 110** BUYERS' GUIDE: Modernise Your Home with Smart LED Bulbs
- 112** SMART LIVING: Gadgets of Tomorrow Revealed at CES 2019

DO-IT-YOURSELF

- 74** Home Automation System for up to Four Devices
- 79** Appliance Guard to Prevent Damage
- 82** 5W Stereo Audio Amplifier
- 84** 5V DC to 48V DC Converter for Phantom Power Supplies
- 86** Minuscule LED Night Marker
- 87** Sending Real-Time Sensor Data through e-Mail
- 89** Portable Emergency Light using Power Bank

Regulars

- 06** Feedback: Your Suggestions
- 08** Q&A: Things You Wanted to Know!
- 10** Tech News: Technology Updates
- 90** Industry News: Industry Updates
- 102** New Products